pfi

Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Case Study: Boosting on Credit Card Default

WEEK 2 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

When the models are this close

Hands-on: 10.1 and 10.2

bootstrap_interval on a validation score, the paired difference, and compare_models with out-of-fold thresholds.

Case study: six models

Fit, tune, and choose inside the training rows; report on the validation rows; describe; say where it fails.

How much of the gap between two models is the models, and how much is which 6,000 accounts were scored?

The test rows stay closed: the case study is where they would usually be opened.

How much of a difference is noise

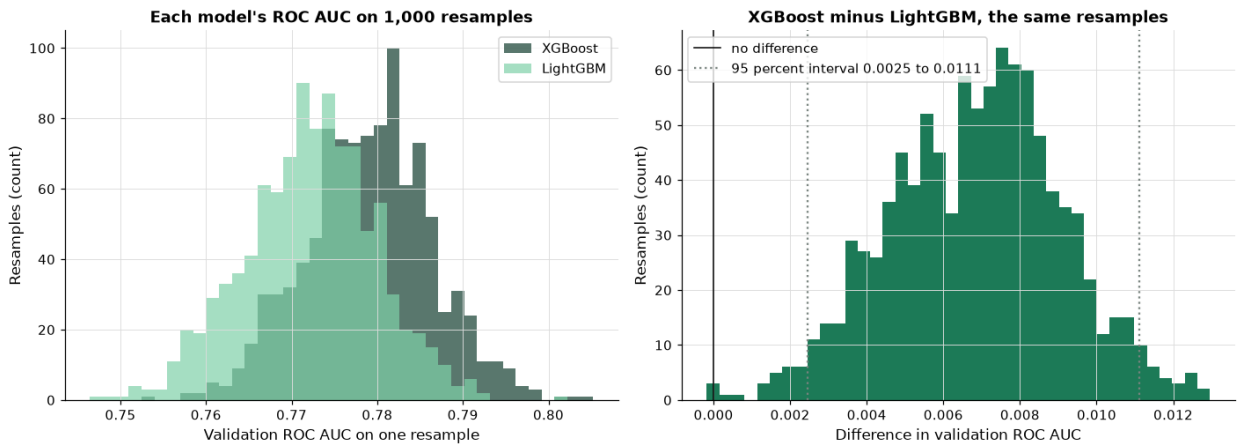
0.7650 to 0.7933

XGBoost, validation ROC AUC 0.7794

0.7572 to 0.7871

LightGBM, validation ROC AUC 0.7725

95 percent intervals from 1,000 resamples of the validation rows; nothing refitted



The paired difference

XGBoost minus LightGBM: 0.0025 to 0.0111. XGBoost ranks higher on 99.9% of the resamples. Both models were scored on the same rows, so they move together.

Thresholds from the training rows

The highest threshold whose out-of-fold recall on the training rows is at least 0.70 (oof_threshold). The validation rows choose nothing.

Validation rows	ROC AUC	Threshold	Recall	Precision
XGBoost	0.7794	0.1812	0.6888	0.4115
LightGBM	0.7725	0.1778	0.6956	0.4043

Recall lands near 0.70, not on it. Moving the threshold now would be choosing on the validation rows.

The case study, test rows closed

18,000 training	fit, tune, choose; 3,982 defaults, 22.12%
6,000 validation	report only; 1,327 defaults
6,000 test	closed: never scored in Course 5

Allowed names

model probability of default in this file

flagged at threshold t

the model's score (weighted or resampled)

Never called

a credit decision

an approval or a decline

a risk grade or credit score

Five models, fitted on the training rows

Model	ROC AUC on its own training rows
Logistic regression (Course 4)	0.7296
Pruned tree (Course 4)	0.7691
Random forest, 200 trees	0.9661
XGBoost	0.8496
LightGBM	0.9220

Pruned tree

ccp_alpha 0.001, chosen by five folds of the training rows: 12 leaves.

These numbers describe the fit. Nothing is chosen on them.

A small search, and what best_score_ is not

Rank	num_leaves	learning_rate	min_child_samples	fold mean	fold sd
1	25	0.0153	333	0.7837	0.0018
2	18	0.0237	368	0.7837	0.0018
3	25	0.0139	150	0.7836	0.0022
4	7	0.0288	392	0.7829	0.0013

```
RandomizedSearchCV(..., n_iter=8, scoring="roc_auc", cv=mlkit.cv_folds(), random_state=SEED, n_jobs=1)
```

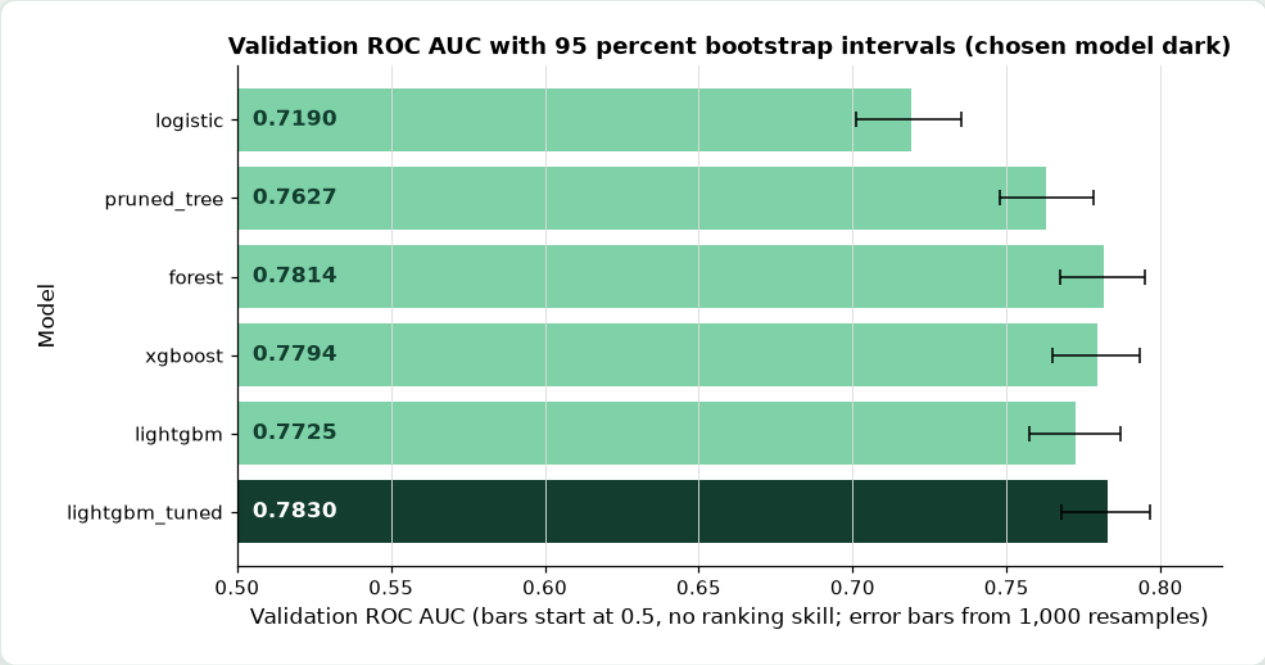
best_score_ 0.7837: the best of 8 noisy fold means, on the folds that chose it. Not an honest score.

The choice, inside the training rows

Model	Threshold	Out-of-fold ROC AUC	Out-of-fold AP
Logistic regression (Course 4)	0.2047	0.7273	0.5084
Pruned tree (Course 4)	0.4015	0.7603	0.5112
Random forest, 200 trees	0.1987	0.7754	0.5503
XGBoost	0.1812	0.7797	0.5542
LightGBM	0.1778	0.7743	0.5483
LightGBM, tuned	0.1890	0.7832	0.5488

Chosen: LightGBM, tuned. Its settings were tuned on the same folds, so its score leans high.

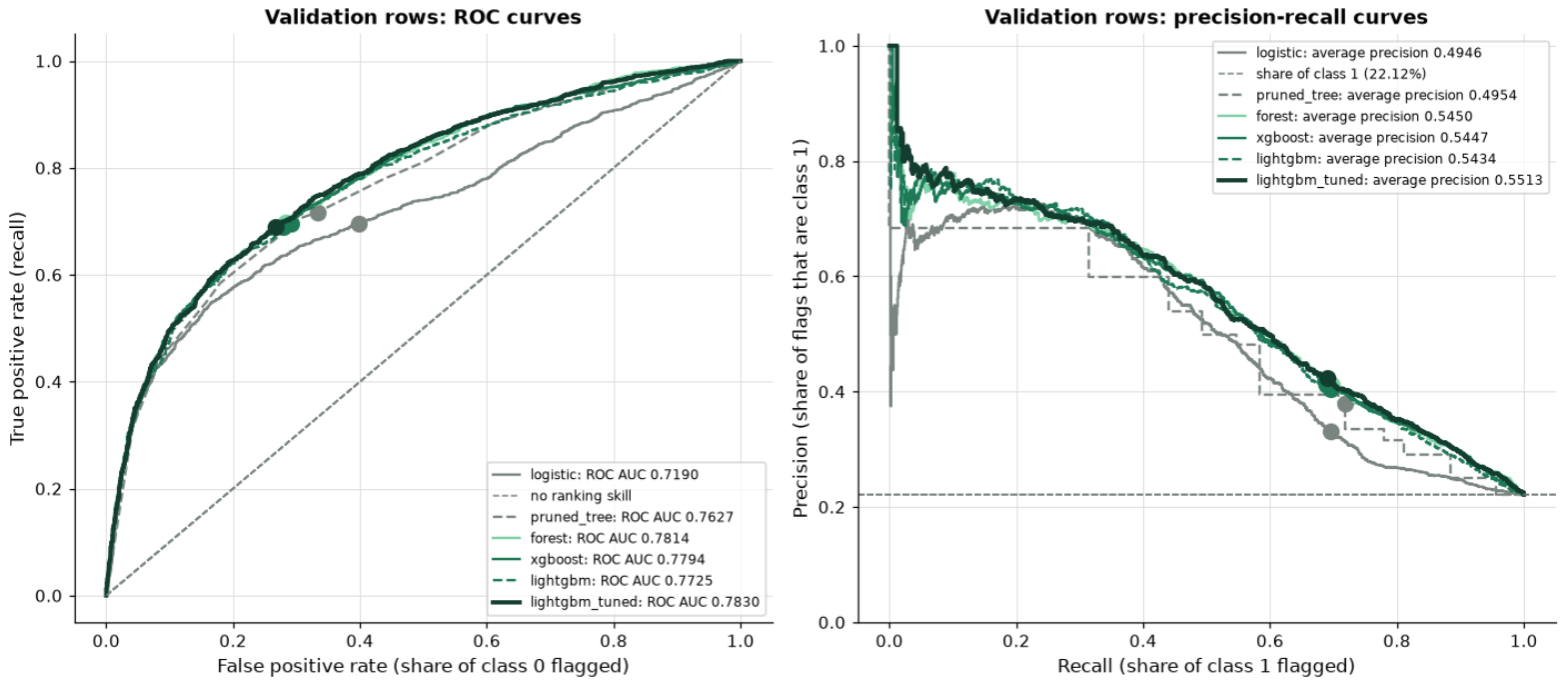
The comparison on validation rows



Validation	ROC AUC	Interval
logistic	0.7190	0.7010 to 0.7352
pruned_tree	0.7627	0.7476 to 0.7784
forest	0.7814	0.7673 to 0.7948
xgboost	0.7794	0.7650 to 0.7933
lightgbm	0.7725	0.7572 to 0.7871
lightgbm_tuned	0.7830	0.7680 to 0.7968

Accuracy sits beside the all-zero baseline (0.7788) and chooses nothing.

Six models, one chart each



The four ensembles' curves lie almost on one line. Each dot is the model's out-of-fold threshold on rows that set nothing.

Paired, against the chosen model

Validation ROC AUC of lightgbm_tuned minus each model, on the same 1,000 resamples

Model	Difference	Paired interval	Chosen ahead on	Reads as
Logistic regression (Course 4)	0.0639	0.0509 to 0.0769	100%	beyond the noise
Pruned tree (Course 4)	0.0203	0.0123 to 0.0279	100%	beyond the noise
Random forest, 200 trees	0.0016	-0.0040 to 0.0070	71%	within the noise
XGBoost	0.0036	-0.0006 to 0.0079	95%	within the noise
LightGBM	0.0105	0.0053 to 0.0153	100%	beyond the noise

No interval here covers refitting with another seed.

What the chosen model leans on

Column	Permutation	Split count	Split rank
PAY_0	0.0906	335	5
LIMIT_BAL	0.0239	660	1
BILL_AMT1	0.0129	565	2

Permutation importance: the drop in validation ROC AUC when the column is shuffled. For LightGBM, feature_importances_ counts splits, not impurity.

Allowed names

permutation importance; split count

the model leans on

association in this file

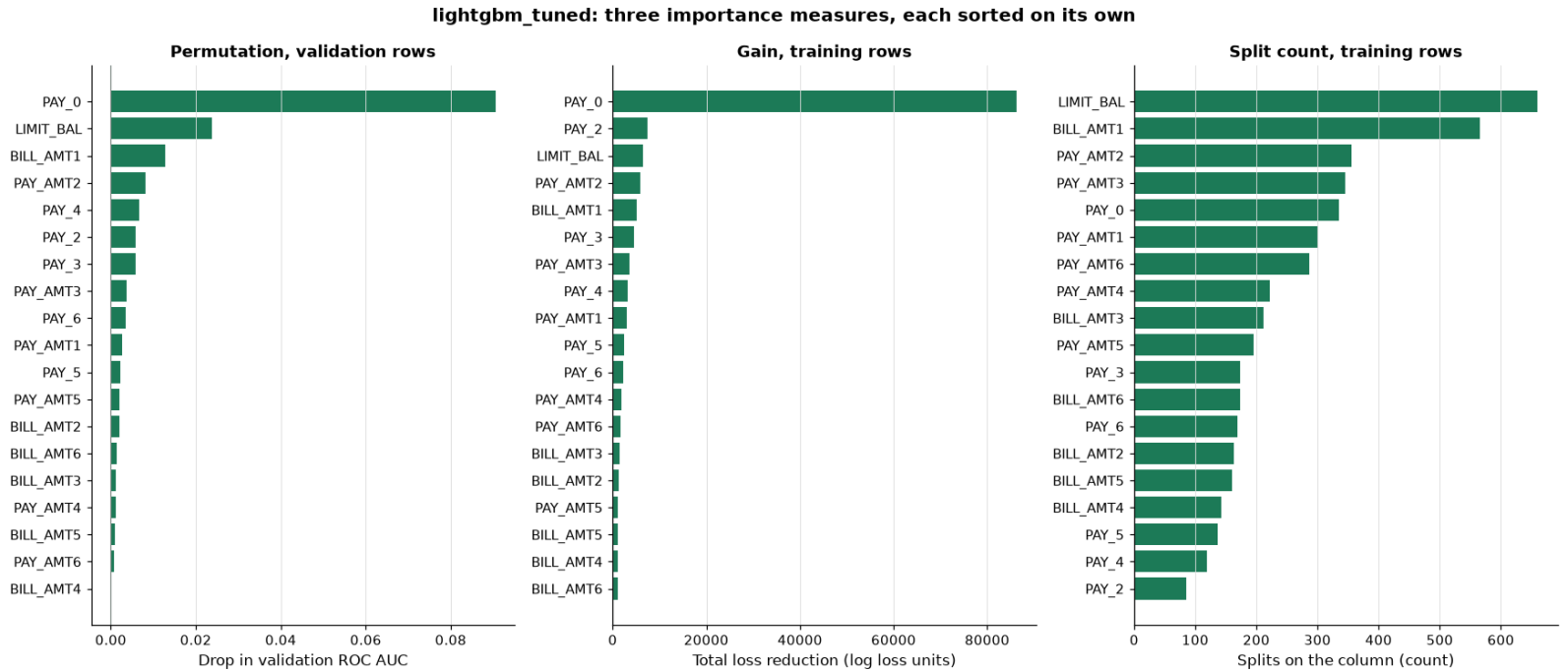
Never called

driver, key driver

cause or determinant

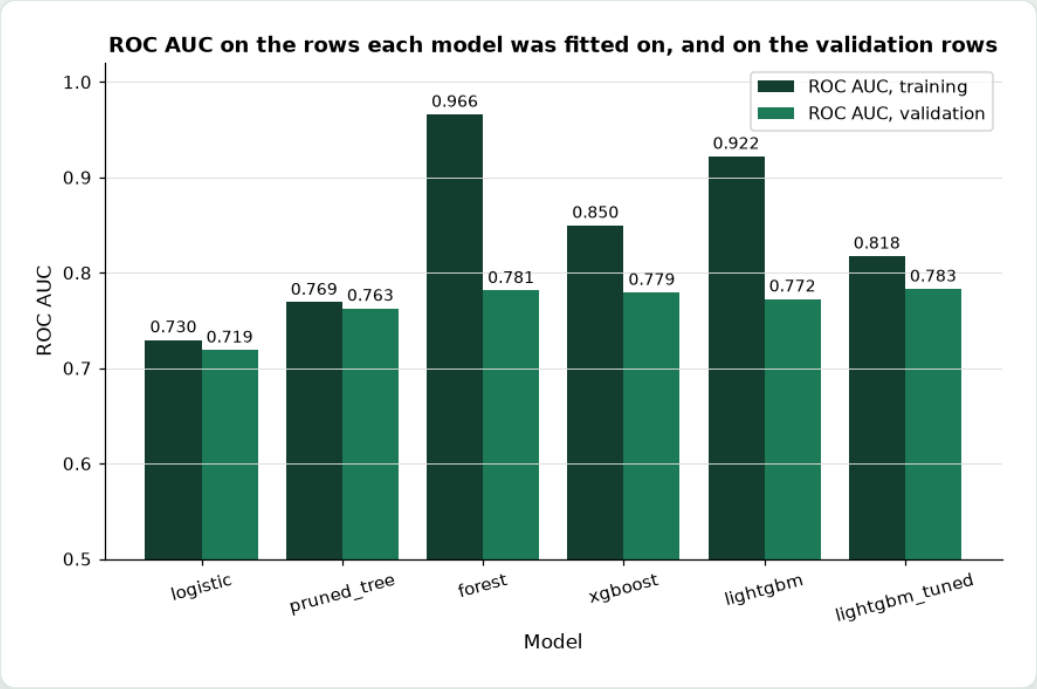
risk factor

Three measures, three orders



Permutation and gain put PAY_0 first, by far. By split count it ranks 5 of 19.

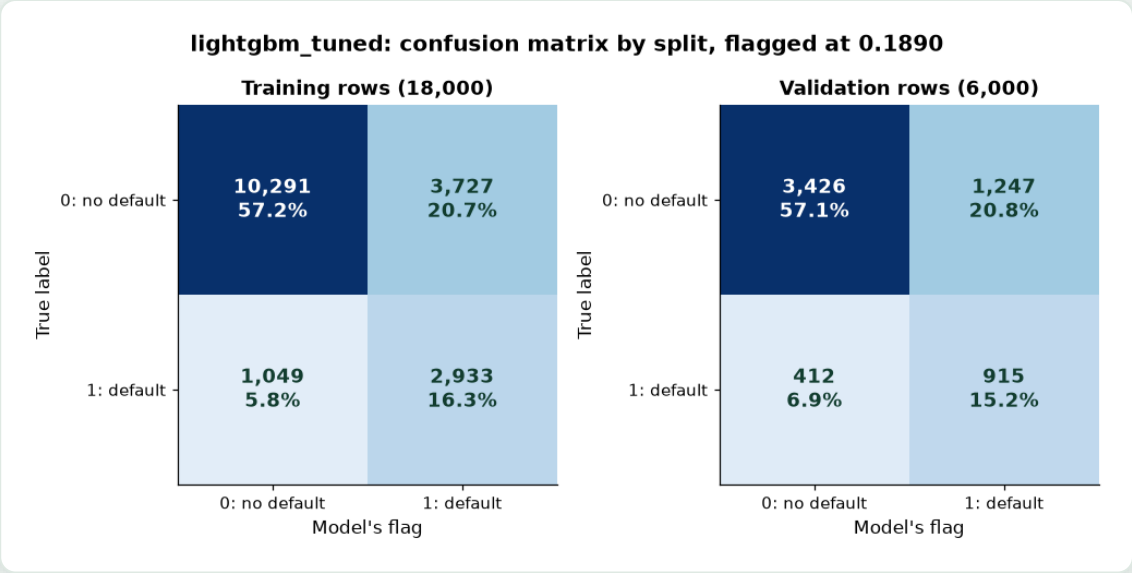
Every model, both splits

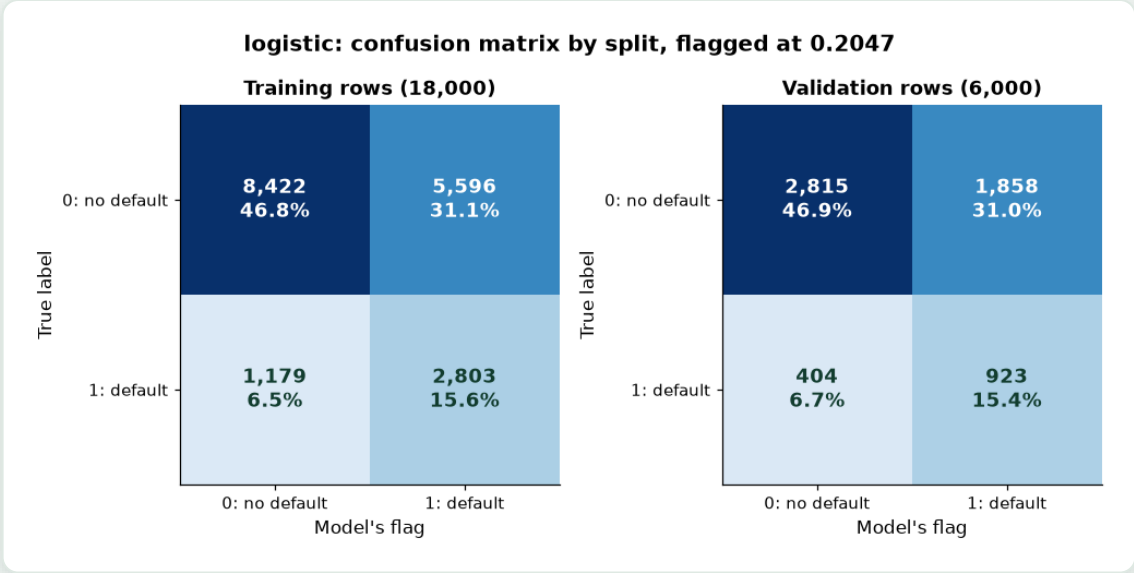


Model	AUC train	AUC valid	Recall train	Recall valid
logistic	0.7296	0.7190	0.7039	0.6956
pruned_tree	0.7691	0.7627	0.7328	0.7174
forest	0.9661	0.7814	0.9957	0.6971
xgboost	0.8496	0.7794	0.7870	0.6888
lightgbm	0.9220	0.7725	0.9214	0.6956
lightgbm_tuned	0.8175	0.7830	0.7366	0.6895

The training columns describe the fit. The widest gap: forest.

The chosen model and the baseline, by split





Training, then validation; each cell is a count and a share of that split's rows. True label down the side, the model's flag across.

What the model shows, and where it fails

403 of 412

missed defaults with a September status of 0 or below

658 of 667

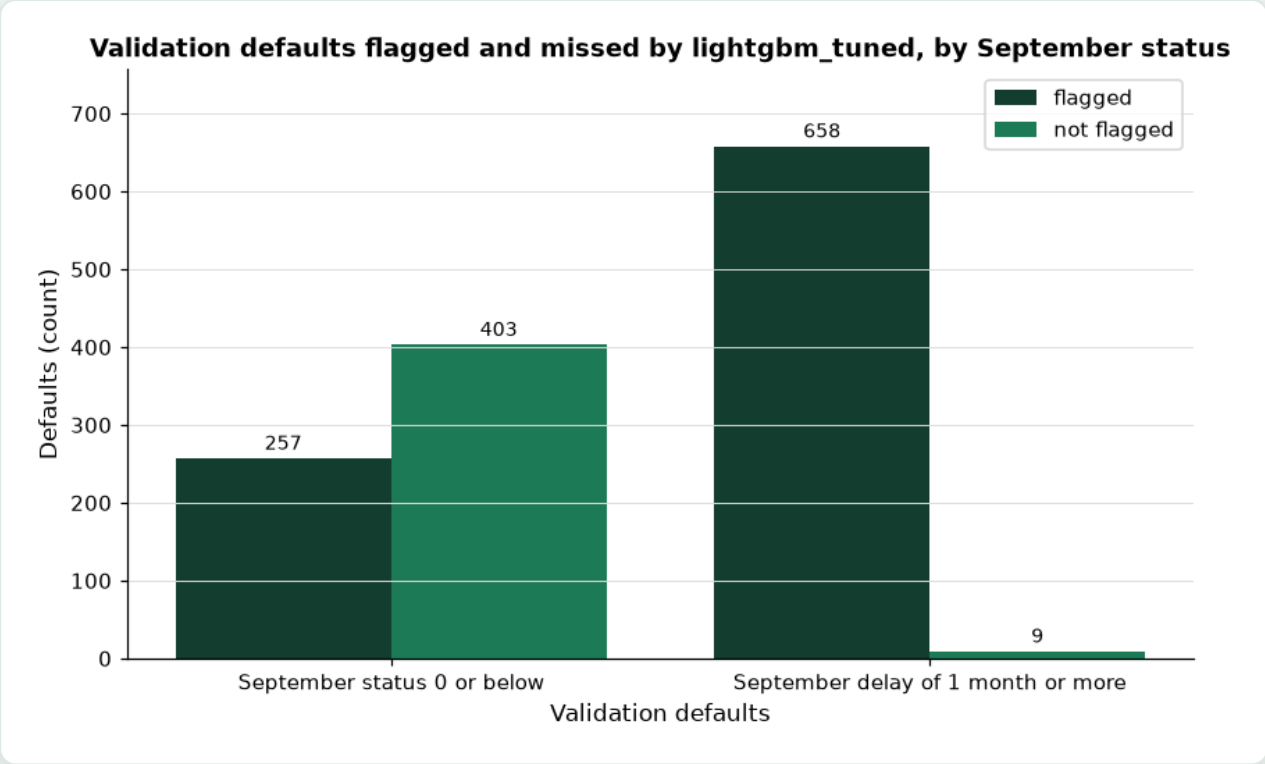
defaults with a recent delay, flagged

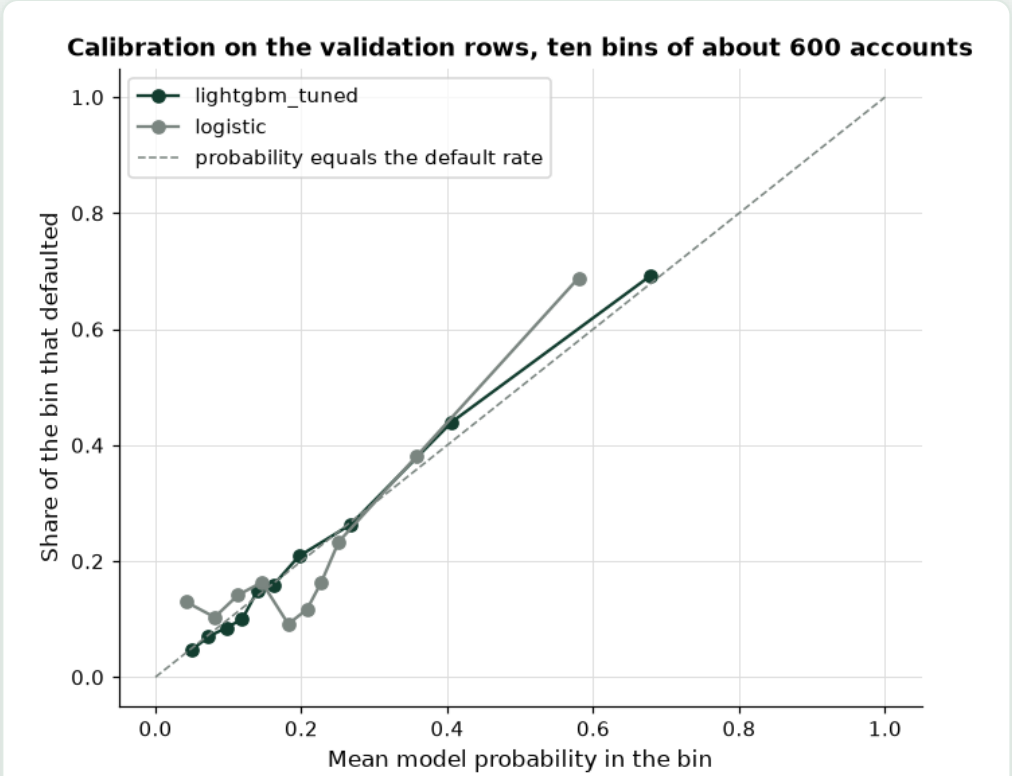
257 of 660

defaults without one, flagged

On rows	594 of 1,247 false flags were late in September and did not default.
On the choice	The paired intervals cannot separate it from Random forest, 200 trees or XGBoost.
On the threshold	Set for recall 0.70 on out-of-fold rows; validation is not a guarantee.
On the data	Taiwan, 2005; two codes undefined; how accounts were chosen is not stated.

Where the misses are, and how the probabilities read





403 of the 412 missed defaults had no recorded September delay. The curve describes the probabilities; nothing corrects them today.

Excel side by side

Excel side by side (tested in Excel)

`=PERCENTILE.INC(A2:A1001,0.025)`	0.7650: XGBoost's interval, lower end
`=PERCENTILE.INC(A2:A1001,0.975)`	0.7933: upper end
`=PERCENTILE.INC(C2:C1001,0.025)`	0.0025: the paired interval, lower end
`=PERCENTILE.INC(C2:C1001,0.975)`	0.0111: upper end
`=COUNTIF(C2:C1001,">0")/COUNT(C2:C1001)`	0.9990: share where XGBoost ranks higher
`=INDEX(A2:A7,MATCH(MAX(C2:C7),C2:C7,0))`	lightgbm_tuned: the model 10.7 chose

No forest, booster, search, or seeded bootstrap in Excel. It reads the CSV and the workbook (sheets comparison, intervals, thresholds); every number within 1e-12.

Save the day: closing week 2

1. Week 1

files cut back to week 1, committed, tag week1

2. Today

today's module and tests

3. Log

results.csv: six cv rows, six validation rows

4. Tag

git tag week2

```
git diff --stat week1 week2
3 files changed, 356 insertions(+)
```

38 tests pass at the end of today. A tag works anywhere a commit does.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows choose, which rows report: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a new train_test_split calls 6,000 rows fresh: 1,195 are closed test rows, and the ROC AUC (0.7903) looks better.

Code that runs is not the same as code that is right.

What you can do now

- Put a bootstrap interval on a validation score
- Compare two models by the paired difference
- Choose everything inside the training rows
- Say why best_score_ is not an honest score
- Run a case study with the test rows closed

Week 3, Day 1 Randomized search and honest tuning